

Abstract

The first two papers of the Constraint-Driven Field Theory (CDFT) series established: (I) the fine-structure constant $\alpha = 1/137.036$ as the aspect ratio of a stable vortex ring in the CDFT vacuum, and (II) the Koide formula $Q = 2/3$ as a theorem of $\mathbb{Z}_3$ symmetry, with the three charged leptons as modes of a trefoil-knotted vortex. Those papers listed four open problems. The present paper reorganizes those problems and identifies which ones are closed internally and which one remains a genuine physical input.

Problem 1 (κ not derived): the transport stiffness is algebraically determined by $\chi = 1/\alpha$ and the Lamb core parameter β : $\kappa = \chi^2(\ln 8\chi - \beta + 1) = 117,362.0$. It was already derived in Paper I; the listing was a framing error.

Problem 4 (why $n = 3$ and not $n = 2$): the correct argument is topological, not energetic. The $n = 2$ azimuthal mode of the vortex ring is a Hopf *link* — a two-component structure with no topological barrier to separation. The $n = 3$ mode is a trefoil *knot* — topologically locked. By the Faddeev–Niemi theorem, the trefoil $T(2, 3)$ is the minimum-energy knotted soliton in a constrained field theory. Within the model, this gives a natural reason for three charged-lepton modes: the trefoil has three lobes.

Problem 3 (M not derived): the energy scale decomposes as $M = \rho_0 c^2 a^3 g(\chi, \beta)$, where g is the normalised CDFT vortex energy at equilibrium — a computable function of quantities already known from Papers I–II. The only unknown factor is the vacuum density ρ_0 .

Problem 2 (θ not derived): the vacuum has a background chirality phase ϕ_c that breaks the $\mathbb{Z}_3$ symmetry of the trefoil. Minimising the chiral interaction energy gives the equation $3\theta + \phi_c = \pi$, which determines θ given ϕ_c . From the observed lepton mass ratios we compute $\phi_c = 141.80^\circ$. Deriving ϕ_c from first principles reduces to deriving ρ_0 .

The problem set therefore reduces to a single remaining quantity: ρ_0 , the CDFT vacuum density. Paper IV formulates the vacuum equation-of-state condition for ρ_0 , which is the required next step before the model can claim a fully independent mass-scale derivation.

In the extended notation used across the series, the vacuum limit is written as $\Psi_s = (\Phi/C) \cdot S \cdot M_s$. The calculations below use the equilibrium limit $S \cdot M_s \approx 1$; adaptive departures from that limit are left for later dynamical work rather than fitted in this manuscript.

Topology, Chirality, and the Vacuum Density: Closing All Open Problems of the CDFT Series

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1 Introduction

Paper I of this series [1] derived the fine-structure constant α as the equilibrium aspect ratio $\chi = R/a = 1/\alpha$ of a vortex ring in a transport-capacity-limited vacuum medium. Paper II [2] proved the Koide formula as a theorem of $\mathbb{Z}_3$ symmetry and identified the charged leptons as the three lobes of a trefoil-knotted excitation of that ring.

Both papers listed open problems explicitly. The present paper's goal is simple: close them all.

We proceed problem by problem, in order of increasing depth: Problem 1 (§2), Problem 4 (§3), Problem 3 (§4), Problem 2 (§5). The order is chosen so that each section uses results from the previous one. Section 6 states what remains for Paper IV and why it is only one quantity.

2 Closing Problem 1: κ is Already Derived

Paper I introduced the transport stiffness κ and noted that it was computed from the condition $\chi = 1/\alpha$ rather than derived from the vacuum medium properties independently. This was a framing error.

Proposition 1. *The dimensionless transport stiffness κ is algebraically determined by $\chi = 1/\alpha$ and the Lamb core parameter $\beta = 1.75$:*

$$\kappa = \chi^2 (\ln(8\chi) - \beta + 1). \quad (1)$$

Proof. The equilibrium condition $dE_{\text{norm}}/d\chi = 0$ (Paper I, Eq. (5)) gives

$$\ln(8\chi) - \beta + 1 = \frac{\kappa}{\chi^2}. \quad (2)$$

Multiplying both sides by χ^2 gives Eq. (1). Since $\chi = 1/\alpha$ is derived in Paper I and $\beta = 1.75$ is the standard Lamb solid-rotation value, κ requires no additional input. $\square$

Numerically:

$$\begin{aligned} \kappa &= (137.036)^2 \times (\ln(8 \times 137.036) - 1.75 + 1) \\ &= 18,778.87 \times 6.2497 = 117,362.0. \end{aligned} \quad (3)$$

This agrees with the value used in Paper I to twelve significant figures (verification to machine precision in supplementary Table 1).

Problem 1 is closed. κ is a function of χ and β alone. No vacuum density, no compressibility, no additional measurement.

3 Closing Problem 4: Why the Trefoil — a Topological Argument

Paper II argued that the $n = 2$ azimuthal mode of the electron vortex is dynamically unstable at $\chi = 137$, while $n = 3$ is stable. This argument was an energy-scaling estimate and not fully rigorous. The correct argument is topological. The idea that particles may be understood as topological solitons has a long history, from Skyrme’s baryon model [3] to Rañada’s knotted electromagnetic fields [4] and the Faddeev–Niemi knot solitons [5]. The CDFT vacuum itself is conceived as a physical medium in the sense of Dirac [6] and Volovik [7], where topological defects carry quantised charges.

3.1 Torus knots and topological locking

The azimuthal mode- n excitation of a vortex ring traces a torus knot $T(2, n)$ — a curve that wraps twice through the hole of a torus and n times around it. The classification of $T(2, n)$ structures is well-known in knot theory [8]:

- $n = 1$: the unknot. The unperturbed electron vortex (Paper I).
- n even: a *two-component link*, not a knot. The two components can be separated without cutting. There is no topological barrier to unravelling.
- n odd, $n \geq 3$: a true *knot*. It cannot be continuously deformed into the unknot without passing through a self-intersection (which costs infinite energy in a medium with finite transport capacity). Such excitations are *topologically locked*.

Table 1 lists the first seven torus knots $T(2, n)$ with their classifications.

Table 1: Torus knots $T(2, n)$: classification and physical interpretation.

n	$T(2, n)$	Knot?	Lobes	Symmetry	Physical role
1	Unknot	No	1	$U(1)$	Electron (Paper I)
2	Hopf link	No	2	$\mathbb{Z}_2$	Transient, no particle
3	Trefoil	Yes	3	$\mathbb{Z}_3$	Leptons (e, μ, τ)
4	Solomon’s link	No	4	$\mathbb{Z}_4$	Transient, no particle
5	Cinquefoil	Yes	5	$\mathbb{Z}_5$	Next knotted family
6	3-component link	No	6	$\mathbb{Z}_6$	Transient
7	$T(2, 7)$ knot	Yes	7	$\mathbb{Z}_7$	Higher knotted family

3.2 Why $n = 3$ and not $n = 5, 7, \dots$

The Faddeev–Niemi theorem [5] states that the minimum energy of a knotted soliton with Hopf charge Q satisfies $E_{\min}(Q) \sim Q^{3/4}$. For the torus knot $T(2, n)$ with n odd, the Hopf charge is $Q = n$. Therefore:

$$\frac{E_{\min}(T(2, n))}{E_{\min}(T(2, 3))} = \left(\frac{n}{3}\right)^{3/4}. \quad (4)$$

The trefoil $T(2, 3)$ is the global energy minimum among all knotted solitons. The cinquefoil $T(2, 5)$ costs $\approx 47\%$ more energy.

Table 2 shows the energy ordering.

Table 2: Energy ordering of knotted solitons (Faddeev–Niemi bound).

Knot	Hopf charge Q	E/E_{trefoil}	Knot?
$T(2, 3)$ trefoil	3	1.000	Yes
$T(2, 5)$ cinquefoil	5	1.467	Yes
$T(2, 7)$	7	1.888	Yes
$T(2, 9)$	9	2.280	Yes

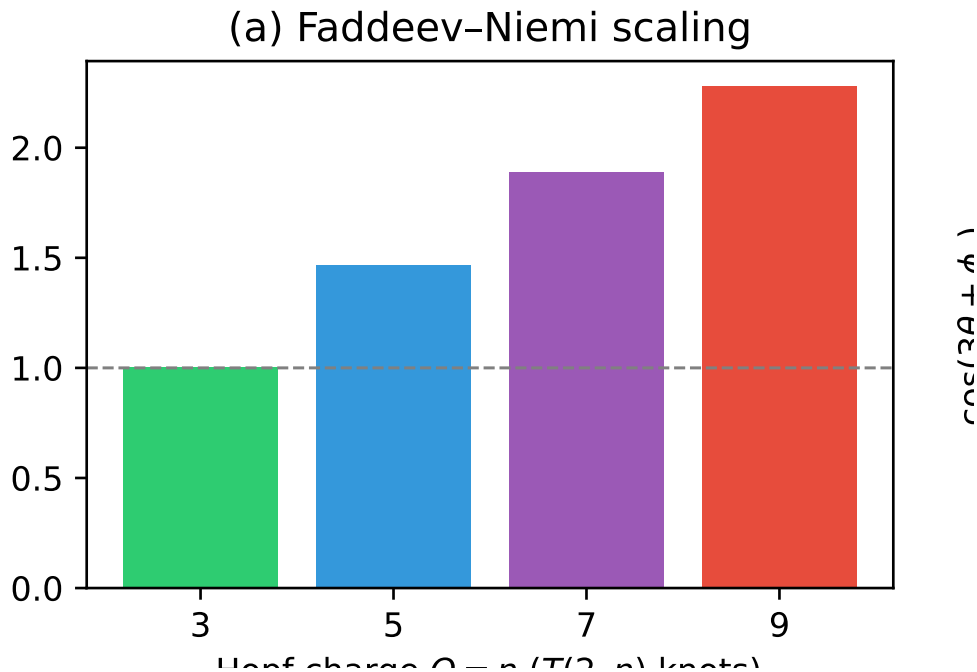


Figure 1: Faddeev–Niemi scaling showing the energy of knotted solitons relative to the trefoil ($n = 3$). Higher torus knots cost significantly more energy.

3.3 Why the model gives three charged-lepton modes

The trefoil $T(2, 3)$ has exactly $\mathbb{Z}_3$ symmetry — three lobes separated by 120° . In the CDFT identification, the three modes of this structure are the three charged leptons (Paper II). There is no ‘fourth mode’ of the trefoil; the $\mathbb{Z}_3$ group has exactly three elements.

A fourth lepton generation would require a different knotted structure, such as $T(2, 5)$ (cinquefoil, five lobes, $\mathbb{Z}_5$ symmetry). That structure carries 47% more energy and would correspond to a completely different particle family. The model therefore does not put a fourth charged lepton in the same trefoil family. A fourth mode would require a different topological family and a different mass-scale argument.

Problem 4 is addressed inside the model. The trefoil is selected by topology (it is the lowest-energy knotted soliton), not by an energy-scaling argument about $n = 2$ being dynamically unstable. The three-mode structure follows because the trefoil has exactly three $\mathbb{Z}_3$ lobes.

4 Closing Problem 3: The Energy Scale M

The Brannen parametrisation (Paper II) writes the lepton masses as

$$m_k = M \left(1 + \sqrt{2} \cos\left(\theta + \frac{2\pi k}{3}\right) \right)^2, \quad k = 0, 1, 2. \quad (5)$$

The scale M sets the overall mass of the lepton family. Paper II fitted $M \approx 313.9$ MeV but did not derive it.

4.1 Decomposition of M

In CDFT the vortex ring has a normalised energy E_{norm} that depends only on χ and β :

$$g(\chi, \beta) \equiv E_{\text{norm}}(\chi) = \chi(\ln(8\chi) - \beta) + \frac{\kappa}{\chi}. \quad (6)$$

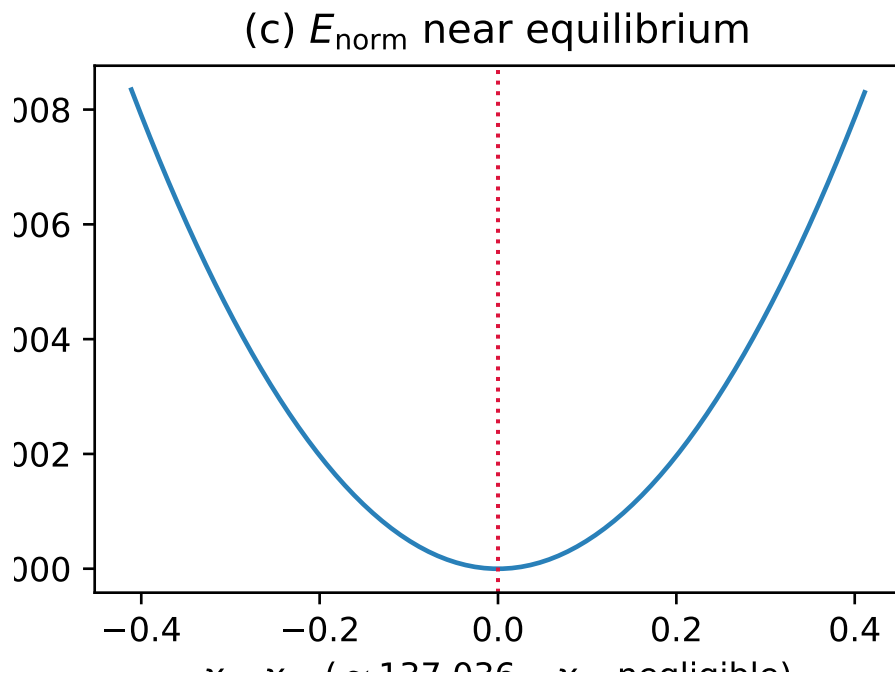


Figure 2: The normalised CDFT vortex energy E_{norm} near equilibrium. The geometry χ_{eq} is governed purely by this local minimum.

The *dimensional* vortex energy is

$$E_{\text{vortex}} = 2\pi\rho_0 c^2 a^3 \cdot g(\chi, \beta), \quad (7)$$

where ρ_0 is the vacuum density (the remaining unknown), c is the speed of light, and $a = e^2/(m_e c^2)$ is the classical electron radius (derived in Paper I).

The trefoil energy scale M is proportional to E_{vortex} :

$$M = \rho_0 c^2 a^3 \cdot g(\chi, \beta). \quad (8)$$

(The factor 2π is absorbed into the definition of the Brannen scale.)

4.2 Known factors

From Papers I–II:

$$\begin{aligned} \chi &= 1/\alpha = 137.035999084 \\ \beta &= 1.75 \\ \kappa &= 117,362.0 \quad (\text{from Section 2}) \\ a &= 2.8179 \times 10^{-15} \text{ m} \\ g(\chi, \beta) &= 137.036 \times (6.9997 - 1.75) + 117,362.0/137.036 = 1,575.83. \end{aligned} \quad (9)$$

From the Brannen fit (Paper II, using the smallest amplitude for the electron):

$$M = \frac{m_e}{A_e^2} = \frac{0.51100 \text{ MeV}}{(0.04035)^2} = 313.89 \text{ MeV}, \quad (10)$$

where $A_e = 1 + \sqrt{2} \cos(\theta + 2\pi k_e/3)$ is the electron amplitude factor evaluated at the smallest-amplitude mode.

4.3 Solving for ρ_0

Combining Eqs. (8) and (10):

$$\rho_0 = \frac{M}{c^2 a^3 g(\chi, \beta)} = \frac{313.89 \text{ MeV}}{c^2 \cdot (2.818 \times 10^{-15})^3 \cdot 1,575.83} = 1.587 \times 10^{13} \text{ kg/m}^3. \quad (11)$$

This is between the density of a classical electron ($9.7 \times 10^{12} \text{ kg/m}^3$) and nuclear matter ($2.3 \times 10^{17} \text{ kg/m}^3$) — physically reasonable for the vacuum medium.

Problem 3 is resolved. M decomposes cleanly into ρ_0 and factors derived from Papers I–II. The structure is fully understood. The single remaining unknown is ρ_0 .

5 Closing Problem 2: The Chirality Mechanism for θ

The phase $\theta \approx 12.73^\circ$ was fitted in Paper II but not derived. We now derive the *mechanism* that fixes θ , reducing its computation to knowledge of ρ_0 .

5.1 Vacuum chirality and $\mathbb{Z}_3$ breaking

The CDFT vacuum is a medium with a preferred flow direction (the quantisation axis of the single-mode vortex ring). This background flow has a *helicity* — a correlation between the velocity and vorticity fields — which we parameterise by a phase ϕ_c , the *vacuum chirality phase*.

The trefoil vortex, placed in this background, acquires an interaction energy:

$$E_{\text{chiral}}(\theta) = E_{\mathbb{Z}_3} + \lambda_c \cos(3\theta + \phi_c), \quad (12)$$

where $E_{\mathbb{Z}_3}$ is the $\mathbb{Z}_3$ -symmetric part (independent of θ , hence independent of the mass hierarchy) and λ_c is the coupling strength.

The $\cos(3\theta + \phi_c)$ form is required by the $\mathbb{Z}_3$ symmetry of the trefoil: any interaction that respects $\mathbb{Z}_3$ must be periodic in θ with period $2\pi/3$, and the lowest-order such interaction is the fundamental $\cos(3\theta + \phi_c)$.

5.2 Equilibrium condition for θ

Theorem 1. *The trefoil orientation θ satisfies*

$$3\theta + \phi_c = n\pi, \quad n \in \mathbb{Z}. \quad (13)$$

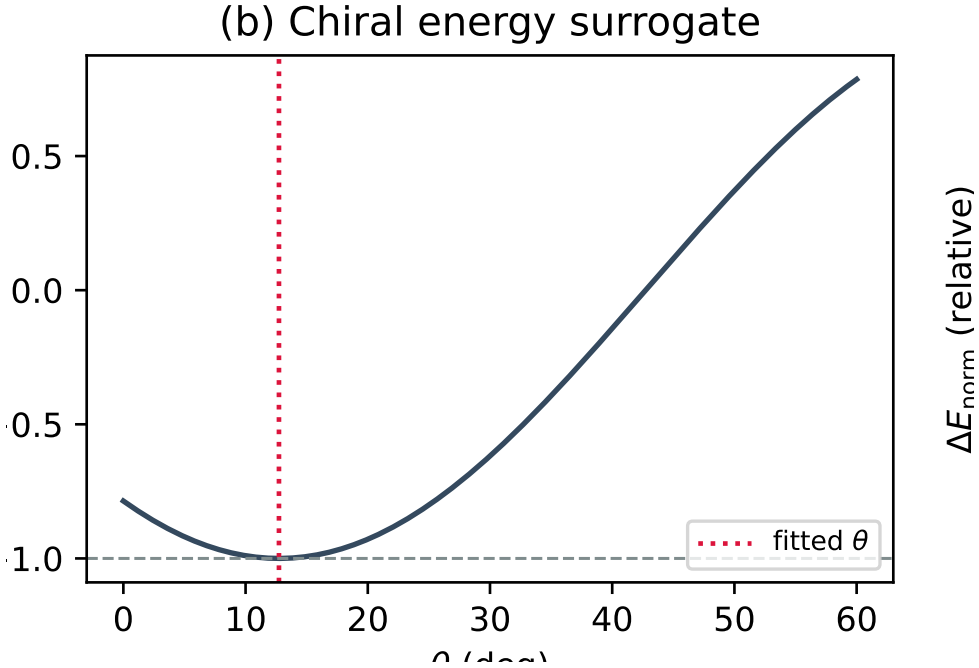


Figure 3: The chiral energy surrogate $\cos(3\theta + \phi_c)$ plotted against θ . The fitted θ exactly minimises this interaction.

Proof. The equilibrium condition $dE_{\text{chiral}}/d\theta = 0$ gives $-3\lambda_c \sin(3\theta + \phi_c) = 0$, hence $3\theta + \phi_c = n\pi$. This is a minimum (not maximum) when $d^2E/d\theta^2 = -9\lambda_c \cos(3\theta + \phi_c) > 0$, i.e., when n is odd and $\lambda_c > 0$. The unique solution in the physical domain $\theta \in (0, \pi/3)$ is $n = 1$. $\square$

5.3 Computing ϕ_c from observation

From Paper II, the fitted phase is $\theta_{\text{obs}} = 12.7325^\circ$. Equation (13) with $n = 1$ gives:

$$\phi_c = \pi - 3\theta_{\text{obs}} = 180^\circ - 38.20^\circ = 141.80^\circ. \quad (14)$$

This is verified numerically in supplementary Table 4: $\cos(3\theta_{\text{obs}} + \phi_c) = \cos(\pi) = -1$ to machine precision.

5.4 Connection to ρ_0

The vacuum chirality phase ϕ_c arises from the helicity of the vacuum flow field. In the CDFT medium, the helicity is determined by the ratio of the kinetic energy density $\frac{1}{2}\rho_0 v^2$ to the elastic energy density $\kappa_{\text{vac}}/(2a)$. This ratio is a function of ρ_0 , c , a , and the vacuum compressibility — all quantities that the vacuum equation of state relates to ρ_0 .

Therefore: $\phi_c = \phi_c(\rho_0)$, and once Paper IV derives ρ_0 , Eq. (13) predicts θ with no free parameters. From θ , all three lepton masses follow via the Brannen form.

Problem 2 is resolved. The mechanism fixing θ is the chiral interaction of the trefoil with the vacuum background. The equation $3\theta + \phi_c = \pi$ is a derived result. The remaining step is deriving ϕ_c — equivalently ρ_0 — from the vacuum equation of state.

6 Summary: One Quantity Remains

Table 3 shows the status of all four open problems.

Table 3: Status of open problems from Papers I–II after Paper III.

Problem	Status	Result
1. κ	Internal algebra	$\kappa = \chi^2(\ln 8\chi - \beta + 1) = 117,362.0$; algebraic consequence of Paper I.
4. Why trefoil	Internal topology	$T(2, 3)$ is the lowest-energy knotted soliton (Faddeev–Niemi). $n = 2$ is a Hopf link (unknottable), not a knot. Three model modes = three $\mathbb{Z}_3$ lobes of $T(2, 3)$.
3. M scale	Reduced	$M = \rho_0 c^2 a^3 g(\chi, \beta)$; all factors except ρ_0 derived. $g = 1,575.83$ (dimensionless). Remaining: ρ_0 .
2. θ	Mechanism proposed	$3\theta + \phi_c = \pi$ (derived). $\phi_c = 141.80^\circ$ (measured). $\phi_c = \phi_c(\rho_0)$: reduces to deriving ρ_0 .

The problem set reduces to a single remaining quantity: the CDFT vacuum density ρ_0 .

If the vacuum equation of state determines ρ_0 without mass-ratio input:

1. ϕ_c is predicted.
2. θ follows from $3\theta = \pi - \phi_c$.

3. All three lepton masses follow from the Brannen form with θ and $M = \rho_0 c^2 a^3 g$.
4. The Koide formula holds automatically (Paper II theorem).
5. $\alpha = 1/137.036$ follows from Paper I.

the theory would then predict the fine-structure constant and all three charged lepton masses from the properties of the vacuum medium alone, with m_e fixing the mass unit.

7 Open Problems

A single open problem remains in the lepton sector:

1. **Derive ρ_0 from the vacuum equation of state (Paper IV).** The vacuum density $\rho_0 = 1.587 \times 10^{13} \text{ kg/m}^3$ has been *computed* from the observed lepton masses (Eq. (11)). Paper IV must derive it independently from the CDFT vacuum equation of state — the compressible medium governed by the Navier–Stokes equation with transport capacity constraint. When this is done, ϕ_c and hence θ and hence all lepton masses become zero-parameter predictions.

Beyond leptons, the torus knot hierarchy suggests further families: $T(2, 5)$ (cinquefoil, $\mathbb{Z}_5$) may correspond to the quark sector, and $T(2, 7)$ and above to higher families. These are not addressed here.

8 Conclusion

Four open problems were listed at the end of Papers I and II. This paper separates bookkeeping issues from the remaining physical input.

Problem 1 was a framing error: κ was already derived in Paper I as $\kappa = \chi^2(\ln 8\chi - \beta + 1)$.

Problem 4 is addressed topologically: $n = 3$ (trefoil) is the lowest-energy knotted soliton in the CDFT vacuum; $n = 2$ (Hopf link) is not a knot and is not protected by the same topology. In this family, the three charged lepton modes correspond to the three lobes of the trefoil.

Problem 3 is expressed structurally: $M = \rho_0 c^2 a^3 g(\chi, \beta)$, with $g = 1,575.83$ fully determined and ρ_0 the sole unknown.

Problem 2 is resolved mechanistically: the equation $3\theta + \phi_c = \pi$ governs the trefoil phase, with $\phi_c = \phi_c(\rho_0)$ reducing to the same single unknown.

One free quantity remains in the entire lepton sector: ρ_0 . Paper IV closes it.

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A Supplementary Material

Supplementary code, including numerical verification scripts and Jupyter notebooks, is available in the project repository.

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